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(1) Research Program/Project/Study Title: Establishing Baseline Agriculture Performance and Development Indicators in the Province of Aklan	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) The main goal of this research project is to establish baseline data on the Department of Agriculture's stakeholders and beneficiaries, their access to and utilization of technology and funding sources, and their feedback on the DA's and its attached agencies' programs, activities, and projects (PAPs) for use in evidence-based planning, policymaking, budgeting, monitoring, and evaluation in the provincial agricultural sector. The study was primarily quantitative in approach and adopted the survey research design. It utilized the farmers, farm workers/laborers, livestock raisers, and fisherfolks who were randomly selected from the sampled municipalities in the province. The study employed Multistage Random Sampling which aimed for representativeness with 400 respondents. Male and female agricultural stakeholders were also represented proportionately. The study utilized a survey questionnaire prepared by the partner agency – UPNCPAG. Results revealed that there are farmers and fisherfolks in the province who are not yet enrolled in the RSBSA despite its implementation in the year 2012. The government's program on seeds and fertilizers distribution was availed by most of the agricultural stakeholders in the province. Despite this, most respondents reported to obtain their planting materials from their own supply. Traditional methods/devices are still mainly practiced in planting, nutrient management, harvesting, and post-harvesting, freezing, and grading. While machinery/equipment is utilized for land preparation. Also, the respondents mostly rely on past experience in the case of fertilization and pesticides use as they are perhaps not updated and are not willing to take risk in their production. The internet is almost unutilized in any agricultural activities in the province. They may not be technologically updated, not capable, and do not have access to internet. In terms of trading, costs of agricultural produced in the province is still prevalently set by the traders/buyers. For those who do not have capital, farmers and fisherfolks would resort to loan to support agricultural and fisheries expenses. The respondents are mostly concern of financial problems rather than health concerns on the first year of the pandemic. Unlike other industries, only very few of them have obtained opportunities during the pandemic (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) farmers, fisherfolks, crops, livestock, fishing, COVID19, financial sources (10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Basically, the Philippines is an agricultural country having bountiful natural resources. Its soils are fertile and tillable and are fitting to various crops and vegetation; and its seas are abundant in both aqua-marine plants and animals. The country's agro-fisheries resources, therefore, can provide a very promising ground for establishing a bedrock of a very strong economy. In the province of Aklan, farming is the basic livelihood in the interior while fishing is the basic livelihood in the coasts. The province posts potentials in the production of high valued and fiber crops. In short, Aklan depends greatly on agriculture. The massive and sustained education and research in agriculture production, the implementation of national program in agriculture, well-established marketing strategies, as well as the support of the agribusiness industry and other private and non-government sectors, result to better production and higher income of the farmers. Palay is still the number one crop grown in the province. Further, fishponds and offshore fishing also provides huge employment for Aklanons. Aquaculture constitutes a significant component in the province's fishery industry. The province has a total fishpond area of 7,807.14 hectares. Aklan is also sufficient in meat and other livestock and poultry products (https://aklan.gov.ph/economy-and-investment/).	

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Despite these ample resources, majority of the local farmers and fisher folks still remain as the sector which falls at the poverty line or even below. Statistics revealed that this sector receives the lowest daily wage compared to their counterparts working in the industry and service sectors (PSA, 2018). This scenario among Filipino farmers is attributed to the insufficient attention given to the agricultural sector (Lubang, 2019). Further, in the case of the fisher folks, they remain poor because they sell their products at a low price, and buy commodities like feedstock at a high price. In short, their expenses are greater than their income (Cooperative Development Authority, 2021).

The Department of Agriculture (DA) is mandated to hit the target of promoting agricultural and rural development through the provision of policy framework, public investments, and support services needed for domestic and export-oriented agricultural enterprises (UPNCPAG, 2021). Despite DA's effort in providing programs and projects aimed at alleviating the lives of the members of the agricultural sector, still a handful of them experience poverty. That is why the study entitled "Establishing Baseline Agriculture Performance and Rural Development Indicators from a Governance Perspective" was initiated by UPNCPAG with the collaboration of state universities and colleges in the country.

In Aklan, CAPSU is involved in the DA Research Project with the primary aim of collecting firsthand data and perspectives from farmers and fisher folks to better understand their economic status and needs from which DA programs can be intensified for the farmers and fisher folks to benefit. Likewise, since there are very limited and not recent studies conducted along this line, it is therefore advantageous that this study was conducted.

The main goal of this research project is to establish baseline data on the Department of Agriculture's stakeholders and beneficiaries, for use in evidence-based planning, policymaking, budgeting, monitoring, and evaluation in the provincial agricultural sector. Specifically, this study had the following objectives:

1. Find out the farmers' and fisherfolks' profile in terms of enrollment in the RSBSA, main livelihood or source of income, land ownership, and affiliation;
2. Determine the DA services/programs availed by the respondents;
3. Identify the respondents' technological utilization;
4. Assess the accessibility and utilization of funding sources for agricultural and fisheries production; and
5. Find out the challenges and opportunities encountered by the respondents during COVID 19 Pandemic.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Research and Sampling Design

The study was primarily quantitative in approach and adopted the survey research design. It utilized the farmers, farm workers/laborers, livestock raisers, and fisherfolks who were randomly selected from the sampled municipalities in the province. The study employed Multistage Random Sampling which aimed for representativeness with 400 respondents. Male and female agricultural stakeholders were also represented proportionately.

Sample Size

The required sample size is set to at least 400 respondents per province. This was the suggested sample size of the DA. Further, 400 is an ideal sample size in a research project with large populations or in a big geographical area.

NOTE: For area – specific coverage – 400 sample size is to be surveyed or interviewed to target a return of $\pm 5.0\%$ margin of error at 95% level of confidence.

Sampling Procedure

Multistage sampling was employed, following three stages. The first stage of sampling was for the selection of sample cities or municipalities. Different cities/municipalities in the province were listed alphabetically and were categorized into two (2) congressional districts (1st and 2nd districts). Random numbers were assigned to

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each city/municipality using the rand () function in excel. These cities/municipalities were then sorted by the random numbers generated. The first five agricultural or coastal municipalities of each congressional district were selected as the sample cities/municipalities. The number of agricultural or coastal municipalities chosen was in proportion with the number of agricultural or coastal municipalities in the congressional districts.

Respondents of the study

To determine the required number of respondents, the designed 400 respondents was divided into the number of congressional districts in the province of Aklan. That is:

$$N \text{ per district} = 400/2 = 200.0$$

That is, the first and second districts were assigned a sample of 200 each. The sample size per congressional district was then divided into the five selected cities/municipalities. That is, sample size per city/municipality is:

$$N \text{ per municipality} = 200/5 = 40$$

Then, divide the sample size per city/municipality into the five selected barangays. That is, sample size per barangay is:

$$N \text{ per barangay} = 40/5 = 8$$

In selecting the respondents for an agricultural area, systematic sampling was employed using the Philippines 2012 data on the Number and the Area of Holdings/Farms by Region, Province and City/Municipality. The respondents were also proportionately chosen by gender.

On one hand, Kish Grid was utilized in identifying the fisher folk respondents having three starting points namely barangay hall, health center, and barangay captain's residence. The researchers made use of the Bureau of Fisheries and Aquatic Resources (BFAR) – Aklan Branch data on the number of fisherfolks in the province. The first sample household is the household on the right of the starting point with two household intervals. The selected respondents were at least 18 years old, and their gender was based on the proportions of male and female fisherfolks according to BFAR's data.

Survey Instrument

The study utilized a survey questionnaire prepared by the partner agency – UPNCPAG. It has undergone review and revisions by the PASUC and the DA. The instrument was approved for use by the UPMREB Ethics Review Panel. The questionnaire is composed of covering letter and nine main sections.

Data Gathering Methods

Data gathering was done through face-to-face interviews which lasted for approximately one hour.

In conducting the survey, two (2) data gathering methodologies were used, which were deemed as the most efficient and effective way of reaching the farmers, farmworkers/laborers, and fisherfolks during the survey fieldwork. The data gathering methodologies are as follows:

1. Face-to-face Interview/Door-to-door interviews. These are most efficient for farmers, farmworkers/laborers, livestock raisers, and fisherfolks who are not always carrying cell phones with them. For coastal areas, the respondents were interviewed via door-to-door. In this method, the respondents were properly identified prior to interview. While for agricultural areas, the respondents were assembled and interviewed in their Barangay Hall.

2. Telephone Interview. Telephone interview was used as a supplemental way of reaching farmers, farmworkers/laborers, livestock raisers, and fisherfolks who are not in their houses or when their location is locked down due to COVID19 during the visit or field survey. In using this methodology, complete contact information of the possible respondents including names and contact details were required.

Data Processing

Field editing by the field interviewers was done every data collection day to ensure completeness of data. All accomplished instruments were submitted to the assigned field supervisor who then checks the completed outputs and final consistency prior to coding of responses by trained coders. Open-ended questions in the questionnaire

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were coded, and code frames were created to facilitate processing of data. Once the questionnaires have been cleared and coded, these were encoded. The data were validated and cleaned prior to data processing.

Statistical Analysis

All data gathered were analyzed using means and percentages.

Ethical Considerations

Risks associated with the study

This study is a socioeconomic and public policy research that employs survey as methodology. There are minimal risks involved in participating in the survey interview. To prevent its spread, COVID-19 health and safety protocols were strictly followed.

Benefits of the Study

There are no anticipated personal and direct benefits to the respondents due to participation in the survey. However, the findings of the study are expected to help improve planning, policymaking, and governance in the Philippine agricultural sector. Data that will be generated from the research will also help in improving the design of government programs and projects related to agriculture. The research is therefore expected to benefit Philippine society in general.

Informed Consent

A Survey Information and Respondent Consent Form (SIRCF) can be found on the first page of the Survey Questionnaire. The interviewers read this form to the respondents and certify that they have done so by signing the form. The SIRCF contains a statement that explains the purpose of the research study and gives an overview of the types of questions that will be asked during the interview. Only prospective respondents who agree participated in the interview. Those who declined were informed that their decision will not affect the services that they could receive or are receiving from the DA or from the CAPSU.

Privacy and Confidentiality

During the survey interview, the respondents were asked for their name and other personal information. However, they are given the option to withhold these details when they choose to do so. Raw data including confidential information were not published and were shared only among the project team.

Table 1. Respondents of the study. Congressional District	Number of Farmers	Number of Fisherfolks	Total	
1st District	160	40	200	
2nd District	192	8	200	
Total	352	48	400	

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

VALIDATION OF THE REGISTRY SYSTEM FOR BASIC SECTORS IN AGRICULTURE

Of the 400 respondents surveyed for this study, 55.75% of them were male while 44.25% were female. Majority of the respondents were married (82.25%), followed by single (9.25%), widowed (7.25%) and were separated (1.25%). The mean age of the respondents is 56 years old. In terms of educational attainment, most (28.0%) of the respondents were high school graduate. This was closely followed by college graduate (21.5%) and some elementary graduate (10.3%), respectively.

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Table 1. Respondent's Profile

Sex	Frequency	Percent
Male	223	55.75%
Female	177	44.25%
TOTAL	400	100.00%

Civil Status	Frequency	Percent
Single	37	9.25
Married	329	82.25
Widowed	29	7.25
Separated	5	1.25
TOTAL	400	100.00%

Age in years	Frequency	Percentage
below 39	61	15.25%
40-49	89	22.25%
50-59	109	27.25%
60-69	97	24.25%
70 and above	44	11.00%
TOTAL	400	100.00%
Mean=56; SD=12.21		

Educational attainment	Frequency	Percentage
None	3	0.75%
Some elementary	58	14.50%
Elementary graduate	41	10.25%
Some high school	44	11.00%
High School graduate	112	28.00%
Vocational	19	4.75%

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Some college	30	7.50%
College Graduate	86	21.50%
Post graduate	7	1.75%
TOTAL	400	100.00%

As to the demographics of people with special status, 32.50% of the respondents were senior citizens, 2.50% were Persons with Disabilities (PWDs), and 2.00% were members of indigenous people (IP) groups or communities. These IP groups were known as Aklanon Bukidnon, Panay Bukidnon, Boracay Tumandok, and Mata IP Group Malay Ati Tribal Association.

The average household size in the Province of Aklan is 4, with 75.50% of the respondents declared themselves as head of the household. The average number of males in the household is 2 while the number of females is 2. The average household members who are currently working in the agriculture and fishery sector (e.g., farmer, farmworker, livestock raiser, fisherfolk) is also 2. The average number of children below the age of 21 years old is 1.0, of which an average of 1.0 is currently attending school.

In terms of access to government services, 97.75% of the respondents have a government ID. Most of the respondents' IDs are PhilHealth ID (32.39%), Senior Citizen's ID (25.4%) and SSS (10.28%).

Barangay officials and employees (96.25%) constitute the most popular source of information on government services, programs, and projects. The average time to reach their municipal or city government offices was reported to be at around 24 minutes.

Table 2. Enrollment in the RSBSA

Response	Frequency	Percentage
1. Yes	313	78.25%
2. No	87	21.75%
Total	400	100%

In terms of coverage of government programs, 361 (78.30%) out of 400 of the respondents, were already registered in the Registry System for Basic Sectors in Agriculture (RSBSA). Among those who are not yet enrolled in the RSBSA, their most cited reason is that they are not informed and they have no idea about the program/RSBSA. On the other hand, 33.50% of the respondents reported to be members of the Social Security System (SSS). Likewise, 22.00% of them also reported that their immediate family is a beneficiary of the Pantawid Familyang Pilipino Program (4Ps).

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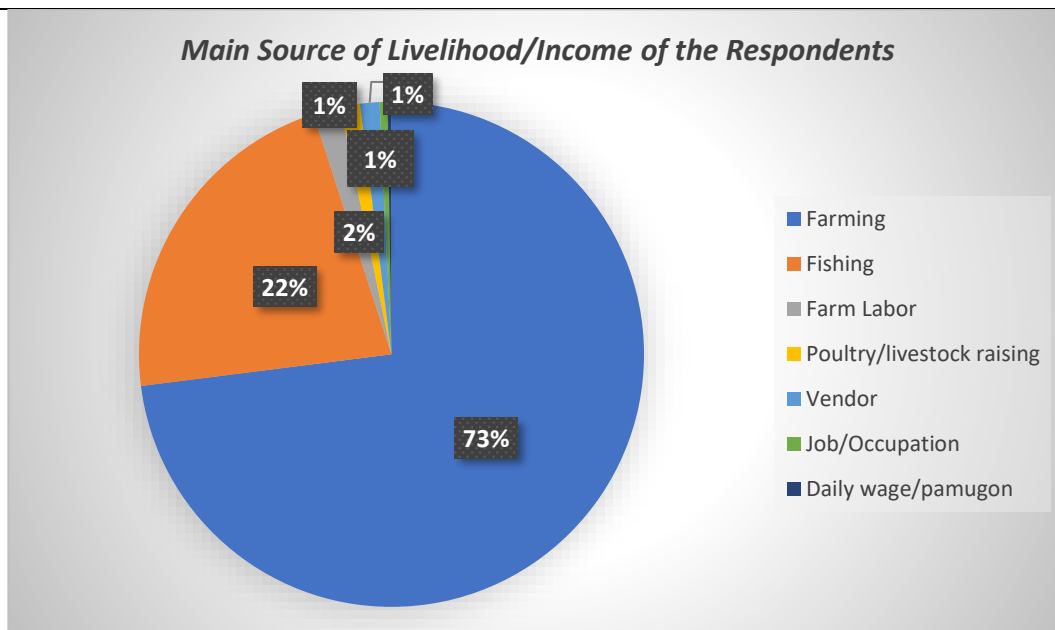


Figure 1. Main Source of Livelihood/Income of the Respondents

Farming was cited as the main livelihood or source of income (73.00%) of the respondents, followed by fishing (22.0%). On the other hand, around 20.50% of the respondents also reported to having received money or remittances from relatives living in another city or province, while 12.25% reported to having received money or remittances from relatives living or working in another country. In terms of land ownership, 67.25% of the respondents own the land where their houses are situated.

On the subject of farmer and fisherfolk affiliation, most of the respondents were found affiliated with any farmer or fisherfolk association or cooperative (62.50%). Some of the most cited reasons for their non-membership is that they are not interested (28.67%), they have no idea (16.67%), they are not informed (9.33%) and mostly some refused to answer why they are not joining associations or cooperatives (40.0%).

IMPACT ASSESSMENT OF DEPARTMENT OF AGRICULTURE (DA) PROGRAMS AND PROJECTS

In order to properly assess the impact of DA-administered programs and projects in the Region, the survey resolved to identify first the leading sources of information on agricultural services offered by the government. Based on the survey results, barangay officials and employees remain to be the primary source of information (99.75%).

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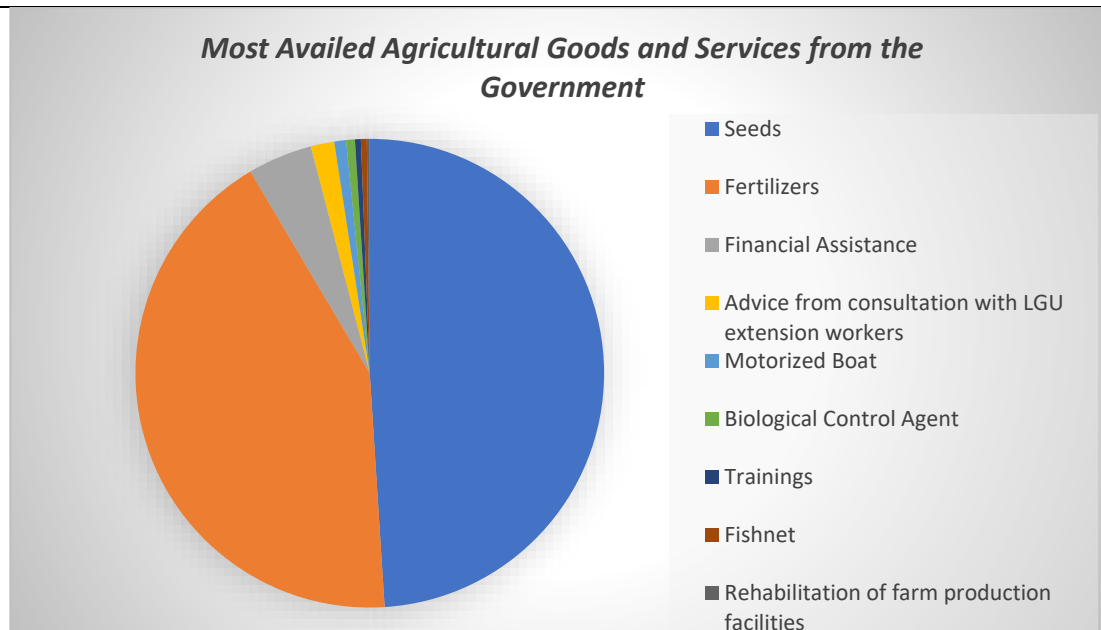


Figure 2. Most Aailed Agricultural Goods and Services from the Government

Seeds (60.5%) and fertilizers (52.5%) were among the most aailed agricultural goods and/or services from the government. The biological control agent, rehabilitation of farm production facilities, trainings, motorized boat, fishnets boats, were the least aailed agricultural goods/and services provided by the public sector.

Few respondents (1.75%) reported to having received agricultural services from NGOs where seeds and fertilizers were the aailed agricultural goods and/or services.

TECHNOLOGY UTILIZATION

General

On the manner of planting, stocking, and materials selection, most of the respondents obtain their planting materials from their own supply of crops/seeds/seedlings/and/or fingerlings (57.25%), followed by the Department of Agriculture (19.50%), vendors/suppliers (16.50%) while some of them are from government in general (5.25%) and local government (1.50%). Most (61.00%) respondents reported that their planting materials and/or fingerlings and stocks were more high yielding, more resistant to pest (61.00%) and more resistant to harsh climatic conditions (59.50%). Most of the respondents reported that the leading cause that prevents them from using planting/stocking materials that are more high-yielding and more resistant to pests or resistant to harsh climatic conditions is the high cost/lack of capital (70.54%).

In terms of preparation for farming/crop production, livestock and poultry raising, and aquaculture, use of seedbeds (58.20%) and pre-germination/direct seeding (34.41%) was found to be the most usual method of seedling preparation_for crop farmers, while use of hatcheries (2.57%) was the most popular strategy for both poultry and fingerling preparation for poultry and aqua farms.

On the manner of obtaining information on prices and availability of inputs, majority of the respondents (93.89%) do not use the internet to learn about the price and availability of inputs such as planting materials, fertilizers, pesticides, and fingerlings.

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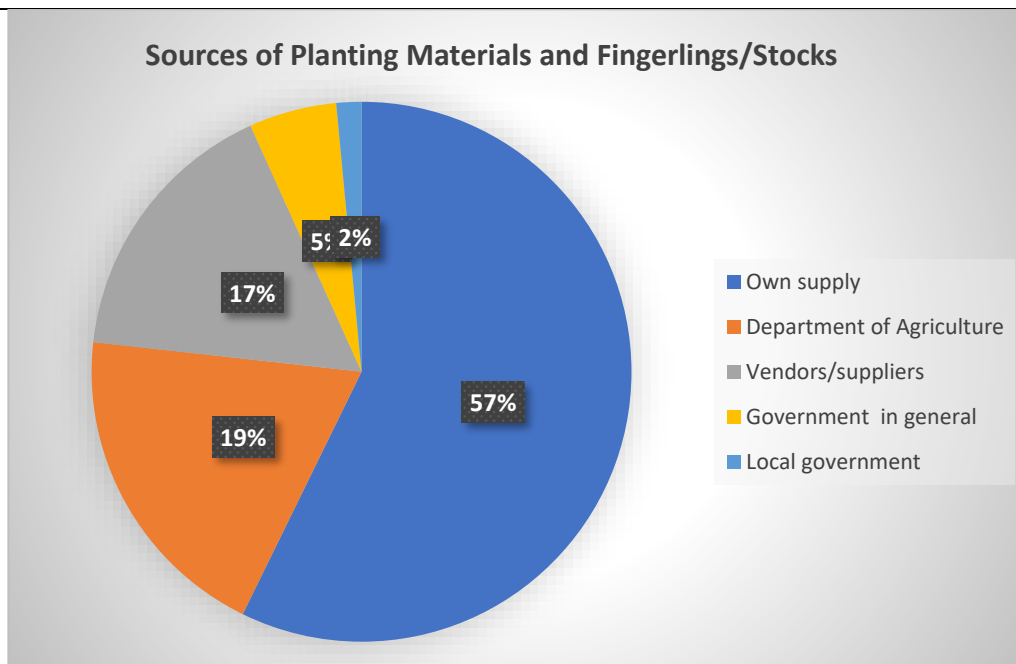


Figure 3. Sources of Planting Materials and Fingerlings/Stocks

Most of the respondents (76.53%) use equipment and technology for production, maintenance, and upkeep of their respective production sites (i.e., farm, livestock/poultry farm, or fish pen/fish cage). Some of the most popular equipment and/or technologies used include hand turtle (46.86%) and hand tractor (39.11%). Manual harvesting (93.75%) was found to be the most popular way of harvesting crops or products among the respondents, while 4.8% reported the use of mechanized equipment. In the case of mechanized harvesting, the leading function is combined harvester.

In the case of agricultural products requiring the use of drying as a method, use of street pavements (62.72%) remain the leading technique employed by the respondents followed by the use of vacant lot (19.00% and multi-purpose pavement (17.20%). For harvests requiring grading or classification prior to selling, majority of the respondents (99.48%) report to having no tool for such purpose. Grading and classification are instead accomplished mainly through visual inspection (100.0%). In the case of products needing refrigeration, styrofoam packaging (93.88%) is the most used technique for freezing.

On the subject of agri-fishery trading, respondents most often use prices identified by the traders or buyers (57.5%) in setting the prices for their harvests or products followed by word of mouth (31.0%), while other respondents (10.5%) do not sell their produced (for own consumption only). Majority (97.5%) do not use the internet to determine the prevailing market prices for their agricultural produce and (97%) also do not use the internet to offer to sell their harvests or products. Around 50.56% of the respondents reported that they transport their harvests or products from their farms to the markets or other places where they could be sold but other buyer or trader pick-up their harvest at the site (49.44%). In such cases, tricycle (39.78%) and motorcycle (34.81%) constitute the most widely used transportation option by the respondents for moving their produce from the farm to the market.

Crop Farming

Around 74.25% of the respondents from the province reported to having earned part of their livelihood from crop farming. Of these, 74.20% are self-reported organic farming practitioners, and 97.70% consider themselves farmers while 2.30% consider themselves as farm laborers. An estimated 44.10% are agrarian reform beneficiaries, while 47.20% are non-beneficiaries. The leading types of work undertaken by the respondents in

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the farm include land preparation (93.98%), harvesting (61.20%), planting and transplanting (48.16%) and cultivation (41.81%).

In terms of tenure status of farmlands, majority of the respondents (50.80%) are owners, while 43.80% are tenants and 4.00% are leaseholders. For the non-landowners, 68.20% of land farmed are owned by immediate family member while 31.80% are owned by a non-immediate family member. Most of the respondents (45.5%) reported to farming lands not covered by agrarian reform and 44.50% are covered by agrarian reform.

The average land area being farmed by the respondents is 1.37 hectares, most of which (52.17%) are rain-fed lowland. Around 28.76% of these lands are irrigated, while around 19.06% are upland rain-fed. Majority of the irrigation is provided by the National Irrigation Administration (67.44%), Other respondents depend on river/sapa (13.95%), water pump (8.14%) and private individual/group (5.81%)

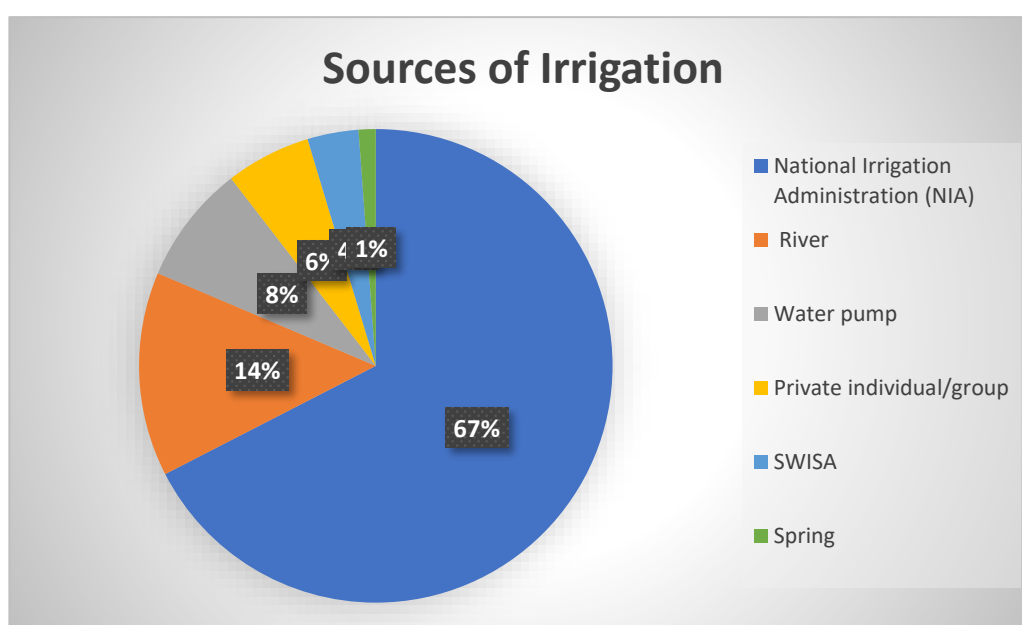


Figure 4. Sources of Irrigation

Rice (94.31%) was the crop grown by most of the respondents in the province in the last 12 months. The average area of land used for rice was 2.72 hectares. The usual seed type used for the said crop is native/traditional (78.60%), while the usual seed source is from government (36.88%), purchased from commercial outlet (28.72%), and combined government and commercial (17.73%). On the other hand, the usual fertilizer type used is inorganic (94.35%), while the usual fertilizer source is the commercial outlets (86.0%). Meanwhile, the usual pesticide type is inorganic (97.30%), while the usual pesticide source is the commercial outlet (96.82%).

In terms of site preparation, most of the respondents use farm animals (41.13%); while others use hand tractors (31.10%) and garden tools (19.06%). In cases wherein machineries or equipment are used for site preparation, an average of 9 days was reported in order to prepare an average plot size of 1.38 hectares for farming. Majority of the functions was land clearing (93.31%) while some perform plowing (3.68%), harrowing (1.34%), land leveling (1.34%), and excavating (0.33%) for the farm needs of the respondents. Meanwhile, around 16.4%% of the respondents report to owning the equipment or machinery used for site preparation.

On the subject of preparation and nutrient management, *sabog tanim* (94.65%) was found to be the most popular method of planting/replanting and transplanting among the respondents. Concerning the use of fertilizers for nutrient management, the survey also revealed that knowledge from past experience (92.30%) was the cited guiding principle by the majority of the respondents in the use of fertilizers. Hand-held or portable sprayers/applicators (89.00%), meanwhile, were the preferred tool, equipment, or technology in applying fertilizers by the majority. Few of the respondents (9.40%) claimed to use organic fertilizers. The average estimated percentage of organic fertilizer use versus commercial and/or synthetic fertilizers is at 40%. For those

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who use organic fertilizers, around 57.14% prepare their own supply of the same. For those who did not, the huge time consumed in preparation (73.43%) was cited as the leading reason for the non-adoption of organic fertilizers followed by the lack of knowledge or information on how to make (15.87%).

In terms of pesticide use, knowledge from past experience (90.60%) was the most cited guiding principle by the respondents. Around 4.30% of the respondents claim to use botanical pesticides, with the application of this type of pesticide estimated at 73.84% of the time versus commercial and/or synthetic fertilizers. Those who did not use botanical pesticides cited time consuming to prepare (70.63%) as one the top causes for the non-adoption of the aforementioned type of pesticide.

Table 3. Farm Productivity for Crops

TOP MAJOR CROPS (Based on the Survey)	Average Quantity	Average Price Per Kilo	Place of Trading and Marketing (Mode)	Average Number of Helpers
1. Rice	35 cavans	15.58/kg	Market	7
2. Corn	7 cavans	18.25/kg	Market	2

Fruit Farming

Around 7.00% of the respondents from the province reported to having earned part of their livelihood from fruit farming. Of these, 7.00% are self-reported organic farming practitioners, and they all consider themselves farmers. An estimated 39.29% have at least one member of the household or are themselves agrarian reform beneficiaries, while 50% are non-beneficiaries. The leading types of work undertaken by the respondents in the farm include land preparation (71.43%) and planting/transplanting (64.29%).

On the subject of tenure status of farmlands, most of the respondents (57.14%) are owners, while 32.14% are tenants. On the subject of land ownership, lands farmed by the respondents were owned by an immediate family member (57.14%) and non-immediate family member (42.86%). Respondents (46.43%) reported to farming lands covered by agrarian reform.

The average land area being farmed by the respondents is 4.7 hectares. Around 75.00% of these lands are upland rain-fed, while 25.00% are lowland rain-fed. In terms of practices used to induce flowering in fruit-bearing trees, regular pruning (50.00%) was cited by the respondents as the most preferred methodology.

Coconut (89.00%) and banana (60.00%) were the most grown fruit trees in the Province in the last 12 months. The average area of land used for coconut was around 1.18 hectares and the average area of land used for banana around 17 hectares. The usual seed type used for the said fruit trees is native/traditional (93%) while the usual seed source of coconut is own supply while for the banana it was purchased from the commercial outlet. On the other hand, the usual fertilizer type used for coconut and banana are both inorganic, while the usual fertilizer source for coconut if from the commercial outlets. Meanwhile, the usual pesticide type for coconut is inorganic/commercial, while the usual pesticides source is the commercial outlet. Further, banana farmers are not using pesticides on the said fruit trees.

On the subject of site preparation, majority (85.71%) of the respondents use garden tools (rake, shovel, and trowel) while 14.29% use hand tractors.

In terms of nutrient management, knowledge from past experience (92.86%) was the most cited guiding principle by the respondents in the use of fertilizers. Most of the respondents (75.00%) do not use tool, equipment, or technology in applying fertilizers

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Majority of the respondents (92.86%) claimed that they are not using organic fertilizers. The average estimated percentage of organic fertilizer use versus commercial and/or synthetic fertilizers is at 25.00%. For those who use organic fertilizers, around 100% prepare their own supply of the same. For those who did not, the lack of knowledge or information on how to make (23.00%) and the huge time consumed in preparation (76.92%) were cited as the reasons for the non-adoption of organic fertilizers.

In terms of pesticide use, knowledge from past experience (89.29%) was the most cited guiding principle by the respondents. Around 3.57% of the respondents claimed to use botanical pesticides, with 100% application of this type of pesticide. Those who did not use botanical pesticides cited time consuming to prepare (85.19%) as one of the top causes for the non-adoption of the aforementioned type of pesticide.

Table 4. Farm Productivity for Fruit Trees

TOP MAJOR FRUITS (Mode)	Average Quantity	Average Price Per Kilo	Place of Trading and Marketing (Mode)	Average Number of Helpers
Banana	2 sacks	13.00/kg	Market	2
Coconut	8.38 sacks	33.28/kg	Market	3

Livestock and Poultry Raising

Around 7.50% of the respondents from the province reported to having earned part of their livelihood from livestock and poultry raising. Pig (70.00%) and chicken (36.67%) constitute the majority of the animals raised or currently being raised in the province in the past 12 months. Intensive livestock farming (70.00%) was found to be the most preferred livestock/poultry farming type, followed by semi-intensive livestock farming (26.70%). Open type with fence (33.30%) and low-cost housing (26.67%) or the use of slatted floor, bamboo and wood, cogon/nipa roofing) was the most widely-adopted animal housing reported by the respondents. Bulugan system (63.3%), on the other hand, was the most preferred method for livestock/poultry propagation.

Table 5. Livestock/Poultry Holding

TOP ANIMAL TYPES (Mode)	TOP BREEDS (Mode)	NUMBER OF HEADS (Average)	INITIAL SOURCE (Mode)
Chicken	Native	59	Own
Duck	Native	37	Commercial
Pig	Hybrid	4	Commercial

The survey also revealed that the respondents who have earned part of their livelihood from livestock and poultry raising have drainage systems (60.00%) in their farms, have feeding racks 46.67%, have lighting systems (10.00%) and have breeding areas (6.70%) and septic tanks (6.70%).

Most government services, meanwhile, was received by the respondents by personal visit to the government offices concerned (83.30%), followed by visits made by government agencies to the farm (16.67%).

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Table 6. Livestock/Poultry Services Received

TOP ANIMAL TYPES (Mode)	SERVICES (Mode)	INSTITUTION / AGENCY (Mode)
Pig	Vaccination	DA/NGO
Pig	Artificial Insemination	BMED Feeds

Fishing Activities

Around 75.00% of the respondents from the Province reported to having earned part of their livelihood from fishing. Fish capture (84.31%) constitutes the most dominant type of fishing activity in the province, followed by aquaculture (13.73%). Most of the respondents (63.00%) engage in fishing more than once a week, while around 29.00% of the respondents engage in fishing everyday. The dominant fishing type is municipal waters (81.00%), followed by marine fisheries (18.00%) and inland or freshwater fisheries (1.00%) and. Most of the respondents reported to have fished mostly in the different barangays near them.

Table 5. Fish/Aquaculture Holding

TOP FISH TYPES (Mode)	VARIETY (Mode)	QUANTITY (Average)	INITIAL SOURCE (Mode)
Salt water	Tulingan Hasa Bangus Manamsi Tamban Danggit Assorted Salmolite Galunggong Pusit Sa-sa Matang baka Bulaw Isdang bato Tagbilawan Baracuda Maya-maya Alloy Bulinao Pasayan Tuna	39 kg	Don't know
Punong/Fishpond/Sangha	Bangus	65 kg	Commercial

Around 76.74% of the respondents reported to be the owners of the fishing boat, while around 20.93% are fishers (i.e., not owner, captain, or master of fishing boat). The modal length of the fishing boat is less than 25 meters (86.05%). Around 18.18% of the respondents received their fishing boats from the government. During the conduct of fishing activities, an average of 3 persons accompanies the respondents in the fishing boat.

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Motorized boats (60.71%) were reported as the most used equipment in fisheries activities, followed by manually paddled boats (39.29%).

Table 6. Fish/Aquaculture Services Received

TOP FISH TYPES (Mode)	SERVICES (Mode)	INSTITUTION / AGENCY (Mode)
Bangus	Feeds	Private
Hasa-hasa, lapu-lapu, bisugo, maya-maya	Fishing boat	DOLE
Tilapia	Fingerlings	DA

The modal type of fish caught by fisherfolk in the province is bangus, hasa-hasa, shrimp, tulingan, bisugo, bulinao, and tamban. The average weight (in tons) of fish caught in each single fishing activity is 90 kilos or 0.09 tons. Meanwhile, majority of the respondents (94.0%) do not engage in the production of seaweeds in the past 12 months. Pollution in production areas (83.3%) was cited as the foremost challenge in seaweed production. Marine products from the river/open sea (84.62%) and pellet feeds (15.38%) was the most used option for feeding fingerlings or stocks. Lastly, respondents cited the ponds (79.92%) and use of fishnets (23.08%) as the most used equipment in aqua farming activities.

SOURCES OF FINANCING FOR AGRI-FISHERIES ACTIVITIES, SAVINGS, AND INVESTMENT PRACTICES

Majority of the respondents (76.53%) reported having adequate capital on their own in the last production season. For those cases of insufficient capital, some of the sources of financing cited by the respondents include loan (72.60%), some of them sell their livestock (1.40%), wait for pension (1.40%), find extra work (1.40%), and gain income in copra (1.40%). Some of the most available sources of agricultural credit cited by the respondents include the locally earning family members/relatives (98.7%), Traders/Buyers/ Consolidators/ Intermediary (98%), Department of Agriculture (97.9%) and Large Farmers/ Landowners (97.8%). Apart from loans, some of the respondents (15.07%) reported that they have received financial assistance from their respective local government units (LGUs) for their agri-fishery activities. Most financial assistance cited is social amelioration (Php5,000).

Despite the existence of such sources of financing in the aforementioned institutions or individuals as recognized by the respondents in the province, the statistics on actual availment from such loan facilities present a different case. Only an estimated 3.04% of the respondents availed of the loan from their locally earning family members/relatives, while 1.28% availed from the Traders/Buyers/ Consolidators/Intermediary and about 0.51% availed from the Department of Agriculture. The most availed loan source was from locally earning friends/ neighbors (10.17%).

On the subject of motivating factors for loan availment, some of the leading factors cited by the respondents include accessibility (61.54%).

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Table 9. Loan Features

Source of Loan	a.Number of loans taken out in the past year	b.Total peso value of the loan	c.Annual interest rate	d.Number of days to process the loan, from application to receiving of the payout	e.Number of documentary requirements
1. Local Gov't Unit (LGU) (province, city/municipality, barangay)	1.3	35285.7	0.2	9.4	3.8
2. Department of Agriculture	1.0	20000.0	1.5	8.0	1.5
3. Other national gov't agencies (e.g., DSWD, DTI, DENR, DAR)	1.0	18800.0	0.1	82.0	2.4
4. Locally earning family members/ relatives	1.1	3500.0	0.2	3.4	None
5. Locally earning friends/ neighbors	1.2	8166.7	0.1	1.0	None
6. OFW friends/ neighbors	1.0	7500.0	0.1	1.0	None
7. Paluwagan (rotating savings and loans associations)	1.5	5000.0	0.1	7.5	2.0
8. Traders/Buyers/ Consolidators/ Intermediary (Middle men/women)	3.0	12957.1	0.1	6.1	None
9. Large Farmers/ Landowners	1.0	15000.0	0.4	2.0	None
10. Informal lender (5-6)	1.5	5000.0	0.1	4.0	1.5
11. Agri/multipurpose/ credit coop	1.0	22666.7	0.1	10.7	9.5
12. Landbank, DBP	1.0	10000.0	0.1	1 day	N/A

Loan Features (cont'd)

Source of Loan	f. Maturity or repayment period	g. Nature of collateral requirements (In kind, e.g., real estate, livestock, and other fixed assets), if required	h.Percent of loan value demanded as collateral, if applicable	i. Loan taken out as individual loan or part of a group and/or through intermediary (2 answers are possible)
1. Local Gov't Unit (LGU) (province, city/municipality, barangay)	14.6	N/A	N/A	Individual
2. Department of Agriculture	4.0	n/a	n/a	Individual
3. Other national gov't agencies (e.g., DSWD, DTI, DENR, DAR)	36.6	N/A	N/A	Individual

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4.	Locally earning family members/ relatives	3.5	N/A	N/A	Individual
5.	Locally earning friends/ neighbors	3.3	N/A	N/A	Individual
6.	OFW friends/ neighbors	Every Cropping/4 Months	N/A	N/A	Individual
7.	Paluwagan (rotating savings and loans associations)	12.0	None	None	Part of a group
8.	Traders/Buyers/ Consolidators/ Intermediary (Middle men/women)	<u>3.3</u>	<u>N/A</u>	<u>N/A</u>	<u>Individual</u>
9.	Large Farmers/ Landowners	4.0	None	None	Individual
10.	Informal lender (5-6)	6.0	N/A	N/A	Individual
11.	Agri/multipurpose/ credit coop	7.5	N/A	N/A	Individual
12.	Landbank, DBP	When there's available money	N/A	N/A	Individual

Purchase of agricultural products (seeds, fertilizer, pesticides, animal feed, etc.) (96.43%) constituted the most cited purpose of loan, followed by personal / household expenses (25%) and purchase inventories/ goods for sale (10.71%). Majority of the respondents were able to repay their loans (90.48%), while some (5.95%) were able to pay in full but with delay.

In terms of crop insurance experience, around 81.50% of the respondents have suffered from typhoon, 74.00% from flood, and 31.00% from pests and diseases in the past 24 months. Most of the respondents (56.27%) cited low production as the impact by calamity to agricultural production.

Some of the respondents (45.80%) who were affected by such calamities reported receipt of relief or assistance from the national government, while 39.80% reported receipt of relief or assistance from the LGUs. Most of the respondents (77.81%) are willing to pay for crop insurance or any other type of insurance. The potential insurance providers most cited by the respondents is the Department of Agriculture (95.45%) while some are from Philippine Crop Insurance (2.07%).

On the other hand, around 46.50% of the respondents have already tried applying for crop insurance. For those who did not opt to apply for crop insurance, some of the reasons cited that they have no time, no idea and not interested to apply. Some of the respondents (31.72%) reported having received money from their crop insurance claim. The average amount released to the beneficiaries was Php4000.00. and most of them (40.68%) received their pay-out in 3 months' time after successful filing of a crop insurance claim.

Meanwhile, some of the coping mechanisms cited by the respondents in response to losses and damages from calamities and other unforeseen events include re-planting (40.50%) and work for a daily wage/ "pamugon" (33.00%). Majority of the respondents (83.50%) claimed to be willing to group together with fellow farmers/livestock raisers/fisherfolk in order to make financial support more available.

Most of the respondents (85.25%) declared not having been able to set aside part of their income as savings. For those who have set aside savings of their income, securing them in their homes (77.97%) was the most popular manner of safekeeping savings among the respondents, followed by through a family member for safekeeping (16.95%) and in a savings bank account (5.08%). Rural banks and Land Bank of the Philippines were the most popular banks for the respondent's bank savings.

On the other hand, some (23.5%) of the respondents claim to have set aside portion of their income as investments. Household appliances (63.1%) was the preferred investment, followed by Equipment/vehicle (15.4%), and motorcycle (8.5%).

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AGRICULTURAL IMPACTS OF THE COVID-19 PANDEMIC AND COPING MECHANISMS OF STAKEHOLDERS

Disruption in farming, livestock raising, or fishing activities (76.25%) and restrictions on marketing agricultural produce (26.50%) were the two most commonly cited challenges that were encountered by farmers, fisherfolks, and livestock raisers during the COVID-19 pandemic. Most of the respondents (83.3%) reported having received relief/social amelioration during the pandemic. Only 5.75% of the respondents reported having new livelihood opportunities during the health crisis. Among respondents who were able to obtain new opportunities during the pandemic stated that they engaged in business/buy and sell (30.40%), opening a sari-sari store (21.70%), work in construction (17.4%), gardening (13%) while other became a tanod, mushroom producer, attended trainings and habal-habal driver.

On the other hand, there was no concerns reported by the respondents concerning the COVID-19 pandemic.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSIONS

1. Despite DA's effort, there are still Aklanon famers and fisherfolks who are not enrolled in RSBSA, do not own land, availed fewer goods and services from the government.
2. Traditional methods/devices are still mainly practiced in planting, nutrient management, harvesting, and post-harvesting, freezing, and grading. While machinery/equipment is utilized for land preparation.
3. The respondents mostly rely on past experience in the case of fertilizers and pesticides use.
4. In terms of trading, costs of agricultural produce in the province is still prevalently set by the traders/buyers.
5. Despite agriculture being the major source of livelihood, income from agricultural activities do not often result to investment or savings.
6. Disruption in farming, livestock raising, or fishing activities and restrictions on marketing agricultural produce were the most challenges encountered by farmers, fisherfolks, and livestock raisers during the COVID-19 pandemic.

RECOMMENDATIONS

1. Intensification of information dissemination may be conducted by the DA to increase/complete RSBSA registration and to raise awareness regarding services of the government for farmers and fisherfolks.
2. Programs on improving stakeholders' knowledge on responsible and appropriate use of planting materials/fingerlings, pesticides and fertilizers may be implemented by the agricultural extension workers in the province.
3. The DA together with other concerned agencies may come up with a detailed mechanization plan of the agriculture/fishery industry in the province.
4. Enhanced loan/financial assistance program that considers accessibility and maximum repayments may be planned and implemented by concerned agencies.
5. Sustainable and innovative programs on Agri-trading may be implemented by the DA, in partnership with LGU and SUCs to enhance the trading capacities and income opportunities of the agriculture/fishery stakeholders.

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(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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